

CSS Colors:

-> CSS provides multiple ways to apply colors to elements.

-> Used for:

1. Text color
2. Background color
3. Border color
4. Shadows
5. Gradients

-> There are 6 ways to provide the colors in CSS:

1. By Using Color Name
2. By Using RGB — `rgb()`
3. By Using RGBA — `rgba()`
4. By Using Hexadecimal Values
5. By Using HSL — `hsl()`
6. By Using HSLA — `hsla()`

1. By Using Color Name: CSS has predefined color names.

Example: red, blue, green, etc.

2. By Using RGB — `rgb()`:

RGB stands for:

R → Red

G → Green

B → Blue

Range: 0 - 255

Syntax: `rgb(red, green, blue);`

More Examples:

`color: rgb(0, 0, 255); /* Blue */`

`color: rgb(0, 255, 0); /* Green */`

`color: rgb(255, 255, 0); /* Yellow */`

`color: rgb(0, 0, 0); /* Black */`

`color: rgb(255, 255, 255); /* White */`

3. By Using RGBA — `rgba()`:

-> RGBA is same as RGB but adds:

A → Alpha (Transparency)

Range: 0 - 1

0 -> Fully transparent

1 -> Fully visible

0.5 -> 50% transparent

Syntax: `rgba(red, green, blue, alpha);`

4. By Using Hexadecimal Values:

-> Hexadecimal colors start with: #

Range: 0-9 and A-F or a-f

Example:

`color: #0000ff; /* Blue */`

`color: #00ff00; /* Green */`

`color: #ffffff; /* White */`

```
color: #000000; /* Black */
```

Short Hexadecimal:

#ffffff => #fff

#000000 => #000

5. By Using HSL — hsl():

-> HSL stands for:

H → Hue (deg) -> Represents color degree -> 0 to 360

S → Saturation (%) -> Represents intensity of color -> 0% to 100%

L → Lightness (%) -> Represents brightness -> 0% to 100%

Syntax: hsl(hue, saturation, lightness)

6. By Using HSLA — hsla():

-> HSLA is same as HSL but adds: Alpha → Transparency

Syntax: hsla(hue, saturation, lightness, alpha);